

TECHNICAL ASSESSMENT

Language Translation System: Design Brief

For the role of Natural Language Processing Engineer

ISSUED BY	Lexora AI Pvt. Ltd., Language & Conversational AI Group
ROLE	Natural Language Processing Engineer (Product Engineering)
DOMAIN	Customer Support: Multilingual Ticket & Chat Translation
ASSESSMENT TYPE	Take-home design and implementation exercise
DATE OF ISSUE	30 June 2026
TIME ALLOWED	48 hours from receipt

1. CONTEXT

The candidate is being evaluated for a Natural Language Processing role at Lexora AI, where engineering teams build production-grade multilingual systems for customer support platforms used by global product and platform clients. This assessment presents a problem representative of the work the team undertakes, and is intended to evaluate the candidate's ability to design, implement, and clearly communicate a working translation solution.

2. PROBLEM STATEMENT

Design and implement a Language Translation System that automatically translates customer support tickets, live chat messages, and email correspondence between a customer's native language and the language used by the support agent, while preserving the intent, tone, and technical accuracy of the original message.

3. OBJECTIVE

Given an input support message written in any supported source language and exhibiting one or more of the following characteristics:

- (a) informal or conversational phrasing, including abbreviations, emoji, and typographical errors;
- (b) domain-specific or technical vocabulary, such as product names, error codes, and account terminology; or
- (c) mixed-language ("code-switched") content within a single message,

the system shall produce a corresponding output in the designated target language that is fluent, semantically faithful to the source, and suitable for direct use by a support agent or customer without further manual editing.

4. FUNCTIONAL REQUIREMENTS

4.1 Language Pair Support

The system shall correctly accept and translate text across at least four language pairs, covering both high-resource pairs (e.g., English–French, English–Spanish) and at least one low- or medium-resource Indian language pair (e.g., English–Hindi, English–Tamil).

4.2 Core Architecture

The system shall be built around a Transformer-based encoder–decoder architecture. Candidates are required to demonstrate, and justify in their explanation, the role of each of the following mechanisms within their design:

Self-Attention, for capturing dependencies between tokens within each sequence;

Multi-Head Attention, for jointly attending to information from multiple representation subspaces;

Positional Encoding, to preserve word-order information in the absence of recurrence or convolution;

Transfer Learning, through the use of a pretrained multilingual language model (e.g., a BERT-family encoder or a pretrained encoder–decoder checkpoint) fine-tuned for the translation task.

4.3 Processing Pipeline

The system shall accept an arbitrary input message together with a specified target language, perform source-language detection, tokenisation, translation, and any necessary post-processing, and return a single translated output without requiring additional manual intervention.

4.4 Context and Intent Preservation

The system shall preserve the intent, sentiment, and technical accuracy of the source message. For example, an urgent complaint shall not be translated in a manner that loses its urgency. The system shall also correctly handle domain-specific terms, such as product names and error codes, which shall not be erroneously translated or altered.

4.5 Quality of Output

The output translation shall:

- (a) achieve a measurable standard of translation quality against an established automatic metric (e.g., BLEU, chrF, or COMET);
- (b) read as fluent and natural in the target language; and
- (c) be free of hallucinated content not present in, or reasonably implied by, the source message.

4.6 Generalisation

The system shall not be tuned to, or restricted to, a single dataset or a narrow conversational style. It is expected to perform reasonably on previously unseen support messages drawn from a variety of customers, products, and writing styles.

5. INPUT AND OUTPUT SPECIFICATION

Input	A single customer support message in plain text (UTF-8), together with a specified target language.
Output	The corresponding translated message in plain text (UTF-8), in the specified target language.

6. EVALUATION CRITERIA

Submissions will be assessed against the following criteria. Candidates are advised to address each criterion explicitly in their submission.

Criterion	Description
6.1 Translation Quality	The fluency and adequacy of the translated output, evaluated both qualitatively and using standard quantitative metrics where appropriate.

Criterion	Description
6.2 Robustness	The ability of the system to handle informal, noisy, or mixed-language support text of varying length and severity of error.
6.3 Generalisation	Performance on language pairs, products, and message styles not seen during development.
6.4 Architectural Soundness	The correct and well-justified use of attention mechanisms, encoder–decoder design, and transfer learning from a pretrained model.
6.5 Clarity of Approach	The quality, structure, and correctness of the candidate's explanation of their methodology.

7. SUBMISSION REQUIREMENTS

Candidates are required to submit, in addition to the working system, a short explanation video. The video is a mandatory component of this assessment and shall cover, at a minimum:

- (a) the candidate's overall approach to the problem and the rationale for the chosen architecture, including the encoder–decoder design, the attention mechanism(s) used, and any pretrained model leveraged via transfer learning;
- (b) a clear description of how the system works internally, including language detection, tokenisation, the translation model, and any post-processing; and
- (c) a step-by-step dry run on at least one representative support message, demonstrating how the input is transformed at each stage of the pipeline (e.g., tokenisation, attention, decoding).

Submissions that do not include the explanation video will be considered incomplete.

8. TIMELINE

The assessment shall be completed and submitted within forty-eight (48) hours from the time of issue. Late submissions will not be considered unless an extension has been agreed in writing in advance.

Note to candidate: You are encouraged to use any publicly available libraries, frameworks, or pretrained models (e.g., Hugging Face Transformers, mBERT, mT5, MarianMT), provided that all such usage is disclosed in your explanation. Originality of approach is valued, but so is engineering judgement and clarity of communication. If any requirement in this document is unclear, please request clarification before commencing work.